

Unveiling the galaxy-halo connection with gravitational lensing

Dr. Calum Murray

Project Overview

The aim of my research is to understand at a deep level the connection between galaxies and dark matter halos in which they reside. This will necessitate improving upon the current methods for including systematics within observational inference of cosmological and astrophysical parameters. Using Simulation Based Inference (SBI), I will include observational effects at the catalogue level. Crucially, this work will enhance the reliability of galaxy cluster abundance as a cosmological probe, offering new insights into the nature of dark matter and dark energy. Looking to the future, there is a wealth of cosmological information that we cannot yet confidently model; from both an observational and theoretical perspective. Building more sophisticated inference tools is essential in bridging this gap in our understanding of the Universe.

Scientific Objectives

- Observing the galaxy-halo connection (GHC)
 - Build an SBI framework to forward model the galaxy-halo connection; produce a community tool so that other researchers can benefit from this research; use the UNIONS data to drive a world-leading research programme.
- Cosmology with galaxy clusters
 - Incorporate an understanding of the galaxy-halo connection within the Euclid galaxy cluster cosmology analysis; improve cluster weak lensing mass estimation within Euclid; apply SBI methods to cluster abundance analyses.

Overview and Scientific Context

Modern cosmology provides us with a succinct theory of the past 13.8 billion years of the evolution of our Universe. With a relatively simple model, Λ CDM we can describe a remarkable breadth of observations from the anisotropies of the cosmic microwave background (CMB) emitted in the early Universe to the vast clusters of galaxies which emerge in the late Universe.

The next generation of galaxy surveys, such as Euclid, DESI, LSST and Roman [7] have a huge potential to uncover more on the nature of both dark matter and dark energy and beyond this will certainly allow us to push our cosmological model much further than has been previously possible. One can be optimistic for this endeavour given the tensions which are beginning to arise from current data sets: a tension in the expansion rate of the Universe between the CMB and supernova measurements [11], a tension in the amplitude of matter clustering between the late and early Universe [5] and an apparent indication of a time evolution of dark energy equation of state through a combination of galaxy clustering and supernova data [4].

However, these observed tensions also raise an important question: do we understand sufficiently the impact of galaxy formation on the large-scale structure from which these tensions emerge? Current cosmological analyses exclude small angular scales, despite their rich reservoir of cosmological information, as we cannot accurately model observations at such scales. One of the huge improvements with incoming surveys; such as Euclid, will be a significant increase in the statistical power of scales which we can model. However, we should also seek to understand the small-scales that at the moment are poorly understood. **This will be the major challenge in observational cosmology over the next 10-15 years.**

In order to understand the small scales cosmologically, we must understand in detail how galaxies reside within the cosmic web. **This field is called the galaxy-halo connection (GHC), the intricate relationship between galaxies and their host dark matter halos.** This is a dynamic area of research that is being driven currently by both theoretical and observational work. The leading approach to understand these effects is through N-body numerical simulations to produce dark matter

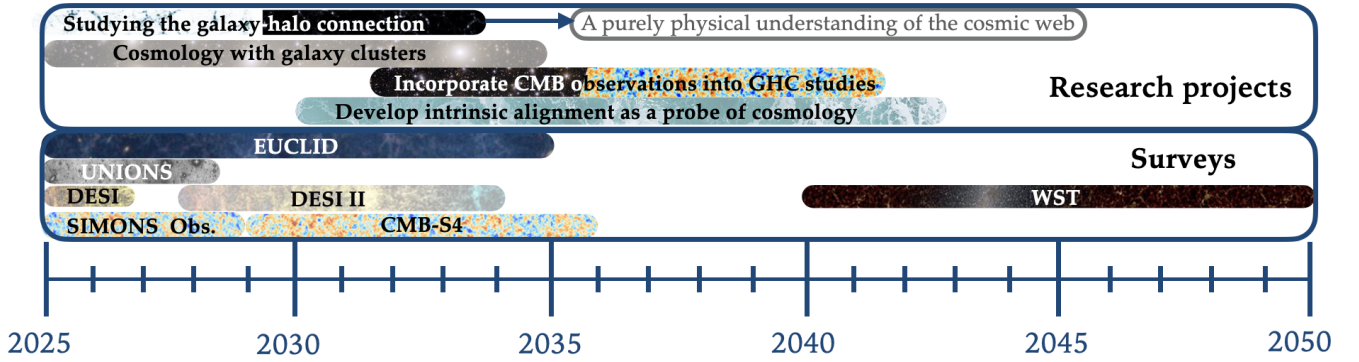


Figure 1: Top: timelines each aspect of my research project. Bottom: surveys with significant French involvement to be used for this research. SIMONS Observatory [9], CMB-S4 [2], DESI and DESI II [3], WST [12], UNIONS [10], Euclid [7].

halo catalogues, followed by statistical prescriptions for galaxy populations within these halos (see [13] for a recent review). In the past comparing theoretical predictions with observations has required simplifying assumptions, such as Gaussian probability distributions, to create tractable likelihood functions for parameter inference. However, these approximations limit the scope and accuracy of the analyses.

The emergence of Simulation-Based Inference (SBI) represents a paradigm shift in this context. SBI enables the direct connection of theoretical predictions with observational data by incorporating observational effects at the galaxy catalogue level. **This approach circumvents the need for restrictive assumptions about likelihood functions, offering a more flexible and comprehensive framework for parameter estimation.** Consequently, the central objective of this research programme in the short and medium term is to develop and apply SBI methods to studies of the GHC, leveraging their ability to bridge the gap between theory and observation. Longer term these methods will help us to extract cosmological information from complex observations; on small scales but also novel cosmological probes such as using intrinsic alignments to constrain the cosmic tidal field.

The AIM laboratory provides an ideal environment for the pursuit of this ambitious research programme. This institute possesses a combination of expertise in galaxy-halo modeling, a observational systematics for photometric surveys, and SBI techniques. Through this project I will strengthen the French contribution to the Euclid mission. France has a world-leading cluster cosmology community, to which I bring my expertise in gravitational lensing and improved methods for cluster mass estimation.

Research programme

The thread between each of my research projects, is the goal of understanding at a deeper level the physical processes underlying the formation of structure in the Universe. For two reasons; to ensure confidence that the observed tensions in cosmological surveys are not due to a mis-modelling of galaxy formation, and to be able to model correctly the Universe across the full range of scales that we observe it. The first project is specifically on the GHC; where I will leverage weak gravitational lensing with new statistical methods and datasets to improve upon past analyses. The second project is on cluster cosmology, which is dependent on a sufficient understanding of the GHC; this project draws upon my extensive experience within cluster cosmology as detailed within my past research statement.

A timeline for the different research projects detailed in this research proposal is shown in Figure 1 alongside the cosmological surveys needed for the research. The short-term priorities are on building upon GHC and galaxy cluster studies, with Euclid, UNIONS and DESI, all surveys with significant input from French researchers. Longer term we can build on understanding intrinsic alignment, which is rich in cosmological information that we cannot yet correctly model and incorporate CMB observations (excellent for baryonic information) into our models for the GHC.

The galaxy-halo connection with lensing: Galaxies trace the underlying matter density field, however we do not yet understand the precise mechanisms governing this relationship. Gravitational lensing, which is sensitive to the entire matter distribution between source and observer, provides a unique opportunity to study this connection in detail. By combining galaxy observations with lensing data, we can disentangle the relationship between galaxies and the matter density field, therefore advancing our understanding of galaxy formation and evolution.

For the distribution of galaxies we require spectroscopic observations and for weak lensing we require galaxy shape measurements with photometric observations. For the next five years, before Euclid observations in the northern sky with its second data release, the best galaxy shape catalogue with complementary spectroscopic observations (from SDSS and DESI) is the Ultraviolet Near Infrared Optical Northern Survey [10] (UNIONS). UNIONS is a 4,861 square degrees photometric survey of the Northern hemisphere which combines observations from three different telescopes (CFHT, Pan-STARRS and Subaru). Importantly UNIONS, with the CFHT r-band, has excellent seeing, making it excellent for the measurement of galaxy shapes. AIM is currently leading the shape measurement of this UNIONS data. **Therefore UNIONS members, such as myself, have a unique opportunity to drive forwards world-leading research on the GHC for the next five years, positioning AIM and myself at the forefront of this field.**

To give a specific example of improvements that can be made to GHC studies with SBI we can look at Figure 2. Starting with a dark matter simulation we produce a galaxy catalogue, following a statistical prescription. From this catalogue we add observational noise and then perform our analysis, precisely as one would with the true observations. Here we look at the weak lensing signal (the shear field) about a pair of central and satellite galaxies. We create a two-dimensional shear map by stacking around the central galaxy position, this process is repeated over many different input parameters for example varying the central and satellite galaxy masses. Then the same procedure is applied to the true observed galaxy catalogue. The true observation is fed into our SBI framework from which we obtain posteriors for our simulation input parameters.

Attempting the same procedure with likelihood-based methods is suboptimal, pixel-to-pixel correlations must be ignored for computational reasons [6] and authors normally reduce the data to a one-dimensional tangential shear field, completely discarding information that is contained in the cross shear field component. This is just one application of SBI to study the GHC. One could then apply this to other interesting relations for example the black-hole halo mass relation, intrinsic alignment and halo mass as a function of the galaxy stellar mass, momentum-field intrinsic alignment correlations and many other observations which are suboptimal with traditional methods.

The expertise required for this project are very similar to those that I have acquired for cluster weak lensing. Within AIM Dr. Martin Kilbinger is an expert in both the GHC and galaxy shape measurement, Dr. Francois Lanusse is an expert in SBI. Dr. Natalia Porqueres, who recently joined the laboratory, is an expert in both the GHC and SBI. Dr. Sandrine Codis is an expert in galaxy formation and more broadly the formation of large-scale structure. Underpinning this research will also be the aim to produce tools for the cosmological community; this will help facilitate reproducible scientific results and add to the impact of my research.

Galaxy cluster cosmology: Building upon the understanding of the GHC developed in the first part of my research programme, I aim to improve cosmological analyses of galaxy clusters and push their analysis to lower-mass scales and therefore recuperate more cosmological information. Euclid will revolutionise galaxy cluster cosmology, increasing the number of detected clusters between redshifts 0.5 and 1.5 by two orders of magnitude (with on the order of 10^4 cluster detections). This redshift range is crucial for probing dark energy, as it spans the epoch during which the Universe transitioned from matter domination to dark energy domination. Of course with this increase in statistical power we must also improve our control of systematics.

As outlined in my research statement, I am excellently positioned to capitalise on these improvements

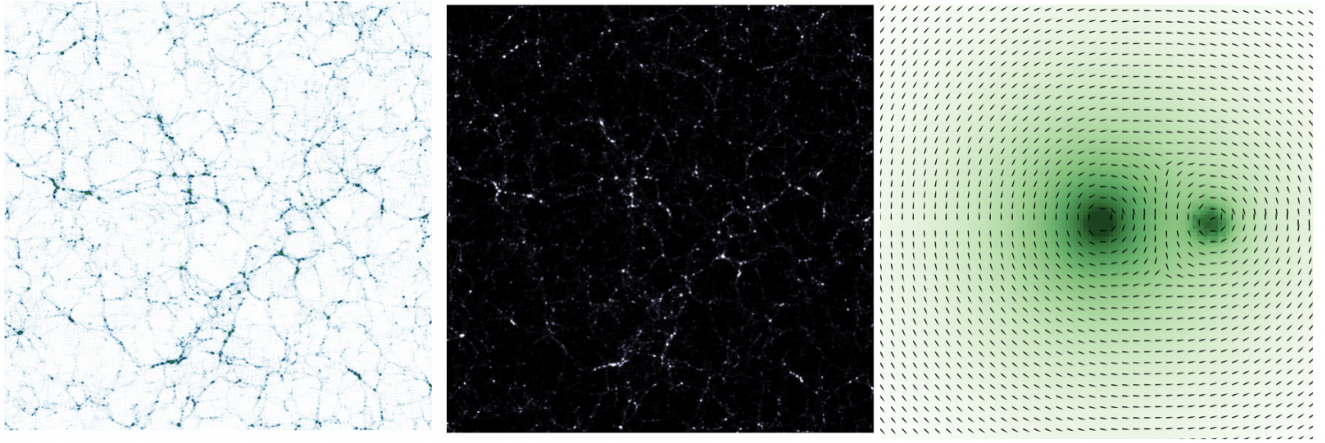


Figure 2: This figure was produced from the Quijote simulations for this report. (a) the dark matter density field (b) an illustration of an optical image using a HOD model on the dark matter density field (c) an example of a simulation and observation output: here the stacked lensing signal around many central galaxy and satellite galaxy pairs.

and contribute to the Euclid galaxy clusters effort, which has strong support from the French cosmology community. The two-dimensional lensing approach discussed in the previous subsection can significantly reduce the noise in cluster mass estimation, a key source of noise in cosmological parameter inference from cluster abundances [8]. Additionally, I will continue developing simulation-based inference (SBI) methods for cluster cosmology, which allows us to use more accurate halo-richness distributions and incorporate complex observational systematics and are therefore well-suited for the demanding analyses required by Euclid.

AIM is an excellent place to lead these efforts, with wealth of expertise in galaxy cluster science. Dr. Marguerite Pierre, an expert in X-ray cluster cosmology, is leading an XMM project which will play a crucial role in characterizing the Euclid cluster selection function¹. Complementing this, AIM host leading experts such as Dr. Gabriel Pratt, an authority on X-ray cluster cosmology, and nearby in Saclay is Dr. Jean-Baptiste Melin, a specialist in Sunyaev-Zel’dovich cluster cosmology. My expertise in optical cluster cosmology and weak gravitational lensing complements excellently the existing expertise of these researchers.

Longer term research goals: The natural progression of this research programme on modelling the GHC through galaxy and weak lensing is to incorporate observations from different physical phenomena. In the near future CMB experiments [9] will provide excellent observations of the movement and density of electrons with Sunyaev-Zel’dovich observations, radio observations will provide us with the distribution of the cosmic plasma through the dispersion of arrival times of different wavelengths from fast-radio bursts and it could soon be possible to map the peculiar velocity field directly using remote CMB temperature measurements².

By incorporating CMB observations into GHC studies we can understand better the role of baryonic processes in galaxy formation. France has a significant involvement in the upcoming SIMONS observatory and CMB-S4. Both of which will provide excellent Sunyaev-Zel’dovich observations on relatively small scales. This is a significant challenge, and one that is currently proving challenging to model [1].

Another challenging observation to model is intrinsic alignment. This is the correlation between the orientations of galaxies across the sky, due to their formation within the cosmic tidal field. Measuring

¹<https://fornax.cosmostat.org/presentation/scientific-goals/>

²<https://indico.in2p3.fr/event/23248/contributions/100126/attachments/66366/92758/Monopole%20fluctuations%20final.pdf>

the tidal field from galaxy shapes will allow us to probe the growth of structure in the Universe. However currently the GHC is not sufficiently understood to use intrinsic alignments for cosmology. By improving our understanding of the GHC we can hope to extract cosmological information from intrinsic alignments. For this research we need spectroscopic galaxy redshifts. In the short to mid-term these can be provided by DESI and DESI II. In the much longer term there will be the wide-field spectroscopic telescope (WST) which is currently under design and of which I am a member [12].

Conclusions

This research programme demonstrates a forward-thinking and innovative approach to understanding the GHC and its implications for cosmology. By leveraging Simulation-Based Inference (SBI) and cutting-edge datasets such as UNIONS and Euclid, I will contribute to a deeper understanding of galaxy formation, cluster cosmology, and the nature of dark matter and dark energy. The proposed work bridges astrophysical and cosmological scales, pushing the boundaries of our theoretical models and statistical techniques while addressing critical challenges in modern cosmology.

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